
Do Agents Learn Pragmatics, or Just Their Partner?

Cross-Play Evaluation of Communicative Self-Play in Situated Reference Games

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Abstract

1 Interactive language agents are often improved or evaluated with the same
2 simulated partners that provide feedback, selection, or reward. In prag-
3 matic communication, this practice can conflate robust listener-adapted
4 language with conventions that only work for a particular partner. We
5 study this issue in PRAG-CrossPlay, a controlled situated reference-game
6 benchmark in which an API speaker generates candidate messages and a
7 listener must identify a hidden target object from a text-rendered scene.
8 Because each listener response is scored against a ground-truth object ID,
9 the task provides an objective communicative-success signal without an
10 LLM judge. We compare mirror self-play, which selects messages against
11 one fixed simulated listener, with population-play, which selects messages
12 against a small heterogeneous listener population, and then evaluate the
13 selected messages with held-out API listeners. Mirror self-play achieves
14 perfect same-partner success but drops under cross-play, especially on
15 perspective-sensitive references. In a 50-scene perspective-stress repli-
16 cation, mirror self-play obtains 1.000 same-play success but only 0.813
17 cross-play success, while population-play reaches 1.000 cross-play suc-
18 cess on the same candidate sets; under gpt-5.5 held-out listeners, mirror
19 reaches 0.673 on perspective stress and 0.653 on partial observability while
20 population-play remains at 1.000. A no-coordinate ablation shows that
21 the strongest population-play result depends on explicit listener-invariant
22 fallbacks and that selector quality matters when those fallbacks are re-
23 moved. The results support cross-play as a simple diagnostic for distin-
24 guishing generally pragmatic messages from partner-specific communica-
25 tive shortcuts.

1 Introduction

26 Pragmatic communication is not only a property of an utterance in isolation. A speaker
27 must choose what to say with respect to a listener’s perceptual context, likely alternatives,
28 prior conventions, and possible misunderstandings. The same phrase can be sufficient
29 for one listener and ambiguous for another: “the small green sphere” may work if there
30 is only one plausible green sphere in the listener’s view, but fail when a second one is
31 present; “the cube on the left” may work for a partner who shares the speaker’s coordinate
32 frame, but become brittle when the listener faces a different direction. These are small
33 examples of a general problem for situated language agents: communicative success is
34 partner-dependent.

35 This partner-dependence matters because recent language-agent systems increasingly use
36 interaction as both an evaluation setting and a learning signal. A simulated partner can
37 answer questions, provide feedback, play the role of a user, or supply a reward through
38 task success. Such interaction is attractive for studying embodied and social language use
39 because it moves beyond static question-answer benchmarks and toward multi-turn com-
40 munication with consequences. But it also creates an evaluation trap. If the partner used
41 during training, filtering, or test-time selection has stable quirks, high interaction reward
42 may mean that the agent has adapted to those quirks rather than learned a robust prag-

44 matic strategy. In other words, an agent can appear socially competent with its training
45 partner while failing with a new partner who interprets ambiguity differently.

46 This concern echoes a long tradition in pragmatics and coordination. Gricean accounts
47 emphasize informative and cooperative utterance choice [Grice \(1975\)](#); collaborative refer-
48 ence work shows that referring expressions are negotiated with particular partners [Clark](#)
49 [& Wilkes-Gibbs \(1986\)](#); [Brennan & Clark \(1996\)](#); and computational reference-game mod-
50 els formalize how speakers should choose utterances relative to listener alternatives [Frank](#)
51 [& Goodman \(2012\)](#); [Goodman & Frank \(2016\)](#). At the same time, multi-agent learning
52 has shown that self-play can produce conventions that are successful internally but brittle
53 under partner shift, motivating cross-play and zero-shot coordination evaluations [Lewis](#)
54 [\(1969\)](#); [Hu et al. \(2020\)](#); [Lucas & Allen \(2022\)](#). Our goal is to connect these views for con-
55 temporary language agents: when communicative success is used as a selection signal,
56 evaluation should ask whether success transfers to new listeners.

57 We study this question in PRAG-CrossPlay, a small, controlled situated reference-game
58 benchmark. A speaker sees a text-rendered grid scene, a hidden target object ID, and the
59 target’s visible attributes. The speaker sends a natural-language message, and a listener
60 who does not know the target ID must choose one object from its visible scene. This set-
61 ting deliberately abstracts away perception in order to isolate pragmatic reference. It also
62 avoids a common source of ambiguity in social-agent evaluation: success is not assigned
63 by a model judge, but by checking whether the listener’s chosen object ID matches the
64 ground-truth target.

65 The key experimental distinction is between *same-play* and *cross-play*. Same-play mea-
66 sures whether a selected message works with the simulated listener used for selection.
67 Cross-play measures whether that same message works with held-out listener models
68 and prompts. We compare a mirror self-play selector, which chooses the shortest candi-
69 date that succeeds with one fixed simulated listener, to a population-play selector, which
70 chooses candidates that succeed across a small heterogeneous population of local listeners.
71 We do not claim that this is a full-scale training procedure. Rather, it is a minimal and
72 inspectable instantiation of the broader interaction-learning pattern: a speaker proposes
73 messages, an interaction partner supplies a communicative-success signal, and the system
74 keeps the message that appears to work.

75 The paper makes three contributions. First, we introduce PRAG-CrossPlay, a released
76 benchmark and protocol with procedural scenes, prompt templates, cached API traces,
77 and verification scripts. Second, we propose a cross-play evaluation protocol for situated
78 natural-language reference that separates same-partner success from held-out communica-
79 tive success. Third, we provide an empirical diagnosis of partner overfitting in API-based
80 reference games. Mirror self-play can reach perfect same-play success while failing un-
81 der held-out listeners; population-play closes the observed gap when listener-invariant
82 fallbacks are present; and no-coordinate ablations show that cross-play also exposes weak-
83 nesses in the selector itself.

84 The main empirical picture is intentionally narrow but, we argue, useful. Robust messages
85 are usually available in the candidate pool: oracle selection often reaches perfect held-out
86 success. The failures arise because mirror self-play selects under-informative or frame-
87 sensitive messages that satisfy its fixed partner. This makes PRAG-CrossPlay a compact
88 testbed for a broader claim: communicative success should not be reported only with the
89 same partner that produced the feedback. For situated language agents, cross-partner gen-
90 eralization is a core part of pragmatic evaluation.

91 2 Related Work

92 **Pragmatics and referring expression generation.** Pragmatic theories of communication
93 emphasize that speakers choose utterances under assumptions about relevance, informa-
94 tiveness, and listener knowledge [Grice \(1975\)](#). Human reference is also interactive: speak-
95 ers and listeners establish conceptual pacts, adapt lexical choices, and repair ambiguous de-
96 scriptions over repeated interaction [Clark & Wilkes-Gibbs \(1986\)](#); [Brennan & Clark \(1996\)](#).

In natural-language generation, referring expression generation operationalizes some of these ideas by selecting attributes that distinguish a target from distractors Dale & Reiter (1995). Rational Speech Act and related probabilistic accounts provide a computational view in which pragmatic speakers choose utterances by reasoning about listener inference Frank & Goodman (2012); Goodman & Frank (2016). Neural and grounded variants apply these ideas to images, colors, and reference-like settings Andreas & Klein (2016); Monroe et al. (2017); Cohn-Gordon et al. (2018); Hu et al. (2021). PRAG-CrossPlay follows this tradition, but asks a different evaluation question: does a message selected by one listener model transfer to other listeners?

Emergent communication, self-play, and cross-play. Multi-agent learning has long used self-play and communicative reward to induce conventions or protocols Foerster et al. (2016); Lazaridou et al. (2017). The benefit of self-play is that it supplies dense interaction data without human annotation. The risk is that agents may converge to arbitrary conventions that do not transfer to independently trained partners. Zero-shot coordination work addresses this risk by evaluating agents with unseen partners or by designing objectives that reduce convention overfitting Carroll et al. (2019); Hu et al. (2020); Lucas & Allen (2022). Our work imports this diagnostic into natural-language pragmatics. Instead of measuring reward in a cooperative game, we measure whether a natural-language referring expression lets a held-out listener recover a ground-truth referent.

Interactive language-agent evaluation. Recent social-agent benchmarks evaluate language models in richer role-play and multi-agent settings. SOTOPIA evaluates agents across open-ended social scenarios with goals such as coordination, exchange, and competition Zhou et al. (2023). SOTOPIA- π extends this line to interactive learning of socially intelligent language agents and raises the concern that LLM-based evaluators may overestimate agents trained through interaction Wang et al. (2024). General LLM-as-judge work similarly shows that model-based evaluations can be useful but require care Zheng et al. (2023). PRAG-CrossPlay is narrower by design. It does not aim to measure broad social intelligence; instead, it gives a small objective test of one communicative skill and makes partner overfitting directly measurable.

3 PRAG-CrossPlay Task

Episode structure. Each episode consists of a scene s , a target object t , a speaker message u , and a listener choice $\hat{t}$. The scene contains a small set of objects $O = \{o_1, \dots, o_n\}$. Each object has a color, shape, size, row, column, and object ID. The speaker receives the target ID and the target’s visible attributes so that it knows which object to describe. The listener receives the candidate-object list and the speaker message but not the target ID. The speaker is instructed not to mention object IDs and to use only listener-visible information.

This setup is text-rendered rather than image-rendered. Rows and columns are explicit, and some scenes include speaker and listener orientations. The abstraction lets us isolate linguistic and pragmatic reference without confounding the analysis with visual perception or robotics execution. It also makes every episode auditable: a verifier can check message leakage, listener choice validity, and success against the source scene.

Scenario families. We generate five scenario families. *Unique-attribute* scenes contain targets identifiable by simple attributes. *Distractor-contrast* scenes include another object sharing salient properties with the target, so the speaker must mention a contrastive cue. *Relational-reference* scenes require a description relative to a listener-visible landmark. *Perspective-shift* scenes give speaker and listener different orientations, making left/right and other frame-sensitive relations brittle. *Partial-observability* scenes include a private landmark visible to the speaker but hidden from the listener, testing whether messages avoid unavailable evidence. The paper-facing stress tests emphasize perspective shift and partial observability because pilot results showed ceiling effects in the easier families.

147 **Success and cross-play metrics.** For a listener ℓ , a message succeeds when the listener’s
148 chosen object equals the target:

$$\text{succ}(u, s, t, \ell) = \mathbb{1}[\ell(u, s) = t].$$

149 For each selected message, we report held-out cross-play success averaged over scenes and
150 held-out listener prompts. For self-play methods, we also record same-play success under
151 the simulated listener used for selection. The cross-play gap is

$$\text{gap} = \text{same-play} - \text{cross-play}.$$

152 A positive gap indicates that same-partner success overstates robustness. We addition-
153 ally report oracle candidate success, candidate-pool robustness, selection regret, random-
154 candidate baselines, message length, and listener-disagreement audits. These secondary
155 metrics help distinguish missing-candidate failures from selector failures.

156 4 Methods

157 **Candidate generation.** For each scene, an API speaker generates $K = 4$ candidate utter-
158 ances. The prompt asks for four different message roles: (1) a natural first attempt, (2) a
159 concise attribute-based message, (3) a relational or spatial message using listener-visible
160 landmarks when possible, and (4) an explicit fallback that may use row/column coordi-
161 nates. Each message is limited to at most 25 words. The prompt includes the hidden target
162 ID only so the speaker knows which object to describe; the system instruction forbids men-
163 tioning that ID, and a benchmark integrity audit checks for object-ID leakage. All API calls
164 are cached by prompt hash.

165 This four-candidate design is important for diagnosis. A single generated message would
166 conflate speaker-generation failures with selection failures. With a small candidate set, we
167 can ask whether a robust message was available and whether each selector chose it. The
168 oracle candidate result is therefore an analysis upper bound, not a deployable method.

169 **Held-out listener evaluation.** Held-out listeners receive the listener-visible scene,
170 speaker and listener orientations, and one speaker message. They return a JSON object
171 containing a choice ID and auxiliary fields such as confidence or ambiguity flags. We use
172 multiple listener prompts that differ in tie-breaking and response style, such as careful
173 comparison versus strict last-match behavior under ambiguity. A selected message’s cross-
174 play score is the average success across held-out listener prompts. Because the returned
175 choice ID is checked against the ground-truth target, the held-out API model is used as a
176 listener rather than as an evaluator assigning subjective quality scores.

177 **Local training listeners.** The self-play selectors use deterministic local listeners rather
178 than additional API calls. These local listeners implement different inductive biases:
179 they vary in tie-breaking, how they treat frame-sensitive relations, and how strongly they
180 weight spatial cues. This design keeps the interaction signal cheap and inspectable. It also
181 makes the result conservative in a specific sense: if mirror self-play overfits even to a sim-
182 ple deterministic partner, the same diagnostic may be useful for richer interaction-trained
183 systems.

184 **Baselines and selectors.** The *template* baseline describes the target with visible attributes
185 and exact row/column coordinates. This is an upper-simple baseline for scenes where co-
186 ordinates are listener-invariant. The *direct* baseline uses the first API-generated candidate.
187 *Best-of-K shortest* selects the shortest candidate and controls for the possibility that merely
188 generating multiple candidates improves quality.

189 *Mirror self-play* scores each candidate with one fixed local listener and selects the shortest
190 candidate that the local listener maps to the target. If no candidate succeeds locally, it falls
191 back to the highest-scoring available candidate according to the same listener. This models
192 a minimal same-partner selection loop: keep the utterance that works with the partner you
193 simulated.

Population-play scores each candidate across three heterogeneous local listeners. It selects the candidate maximizing average local success with a small length penalty for ties. The key difference from mirror self-play is not scale, but diversity: a candidate must work across several plausible listener assumptions rather than one partner’s tie-breaking rule.

Informativeness prior is a text-only selector used in no-exact-coordinate ablations. It prefers candidates containing contrastive or spatial cues such as “only,” “leftmost,” “below,” or “next to.” *Consensus+info* first uses local population feedback when a candidate has unanimous local support and otherwise falls back to the informativeness prior. Neither selector uses held-out listener labels.

Oracle candidate evaluates every generated candidate with held-out API listeners and selects the one with highest held-out success. It is reported only to decompose failures: if the oracle succeeds while a method fails, the candidate pool contained a robust message and the failure is selection rather than generation.

5 Experiments

Runs. We report three bounded API experiments plus a selected-message cross-model audit. The mixed pilot contains 50 scenes from the four initial scenario families. The perspective-stress replication contains 50 fresh perspective-shift scenes. The partial-observability support run contains 50 private-landmark scenes. In all runs, the speaker model generates candidates, local selectors choose final messages, and held-out API listeners evaluate those messages. The cross-model audit reuses the same speaker candidates and selected messages, then evaluates them with gpt-4.1-nano, gpt-5.4-nano, and gpt-5.5 held-out listener prompts. We use paired bootstrap intervals over scenes for method differences.

Model versions and reproducibility. The cache records exact response model versions: gpt-5.4-nano-2026-03-17 for the main API runs, gpt-4.1-nano-2025-04-14 for the alternate-model audits, and gpt-5.5-2026-04-23 for the frontier-listener follow-up. The repository includes generated prompt and schema documentation, a token-accounting report over 4,982 cached responses, an integrity audit that checks generated messages and listener choices, and an artifact guide mapping paper claims to source scenes, cached result files, tables, and verifier outputs. As a benchmark-scale artifact sanity check, the release also includes a no-API 600-scene balanced local sweep: mirror self-play has 1.000 same-play but 0.631 cross-play success, while population-play, template, and oracle candidate each reach 1.000.

What is being tested? The experiments do not test whether parameter updates on a language model improve pragmatic reasoning. Instead, they test a smaller but common mechanism: using interaction-derived success to select among possible messages. This mechanism appears in reranking, rejection sampling, self-refinement, and data filtering. If selection against one partner can inflate apparent success in a controlled reference game, then cross-play is a useful diagnostic before moving to more expensive training or richer social environments.

6 Results

Finding 1: same-play success overstates held-out communicative success. Table 1 shows that mirror self-play obtains 1.000 same-play success but 0.907 cross-play success in the mixed pilot. Population-play obtains 1.000 cross-play success on the same scenes. The paired population-minus-mirror difference is 0.093 with a 95% bootstrap interval of [0.027, 0.173].

Method	Cross-play	95% CI	Same-play	Gap
Template	1.000	[1.000, 1.000]	–	–
Direct	0.840	[0.747, 0.920]	–	–
Best-of-K shortest	0.740	[0.627, 0.853]	–	–
Mirror self-play	0.907	[0.833, 0.967]	1.000	0.093
Population-play	1.000	[1.000, 1.000]	1.000	0.000
Oracle candidate	1.000	[1.000, 1.000]	–	–

Table 1: Mixed 50-scene pilot. Mirror self-play has perfect same-play success but lower cross-play success; the gap is concentrated in perspective-shift scenes.

Scenario-level results localize the effect. Unique-attribute and distractor-contrast scenes are at ceiling, relational-reference scenes show a smaller mirror gap, and perspective-shift scenes are the hard case: direct messages and shortest messages often under-specify the target, and mirror self-play selects messages that work for its fixed partner but fail under held-out listeners. This rules out a uniform degradation story; the self-play gap appears where listener perspective and ambiguity matter.

Finding 2: the gap replicates under fresh perspective-shift scenes. Table 2 confirms the pattern in 50 new perspective-shift scenes. Mirror self-play again achieves 1.000 same-play success but only 0.813 cross-play success. Population-play reaches 1.000 cross-play success, matching the oracle candidate upper bound. The paired population-minus-mirror difference is 0.187 with a 95% bootstrap interval of [0.100, 0.287]. The direct baseline is also substantially weaker than population-play, showing that the gain is not simply due to the API speaker already producing a robust first candidate.

Method	Cross-play	95% CI	Same-play	Gap
Template	1.000	[1.000, 1.000]	–	–
Direct	0.713	[0.600, 0.813]	–	–
Best-of-K shortest	0.420	[0.293, 0.540]	–	–
Mirror self-play	0.813	[0.720, 0.900]	1.000	0.187
Population-play	1.000	[1.000, 1.000]	1.000	0.000
Oracle candidate	1.000	[1.000, 1.000]	–	–

Table 2: Perspective-stress replication on 50 new scenes. Population-play matches the oracle candidate upper bound, while mirror self-play retains a substantial cross-play gap.

Finding 3: the effect persists under an alternate held-out model family. Table 3 evaluates the same perspective-stress candidate sets with an alternate held-out model, gpt-4.1-nano. Direct and shortest baselines become weaker, mirror self-play drops to 0.713 cross-play success, and population-play again reaches 1.000. The paired population-minus-mirror difference under this alternate held-out model is 0.287 with a 95% bootstrap interval of [0.193, 0.380]. This audit is not meant to compare model quality; rather, it checks whether the observed partner-overfitting gap survives a held-out listener shift.

Held-out model	Direct	Best-of-K	Mirror	Population	Oracle
gpt-5.4-nano	0.713	0.420	0.813	1.000	1.000
gpt-4.1-nano	0.553	0.367	0.713	1.000	1.000

Table 3: Held-out listener model audit on the same 50 perspective-stress scenes and same API-generated candidates. The partner-overfitting gap persists when held-out listeners use gpt-4.1-nano instead of gpt-5.4-nano.

Table 4 summarizes the broader cross-model audit. The same selected messages are evaluated under three held-out listener families on perspective stress and partial observability. The mirror self-play gap is not repaired by gpt-5.5; in fact, the population-minus-mirror difference is 0.327 on perspective stress and 0.347 on partial observability. Population-play reaches 1.000 in every row. For the selected-message gpt-5.5 rows, this also establishes an oracle ceiling of 1.000 without a new all-candidate listener sweep, because the population-selected message is one of the candidates and already succeeds for every held-out listener.

Setting	Listener	Direct	Mirror	Population	Oracle	Pop.-Mirror
Perspective stress	gpt-5.4-nano	0.713	0.813	1.000	1.000	0.187
Perspective stress	gpt-4.1-nano	0.553	0.713	1.000	1.000	0.287
Perspective stress	gpt-5.5	0.793	0.673	1.000	1.000	0.327
Partial observability	gpt-5.4-nano	0.740	0.667	1.000	1.000	0.333
Partial observability	gpt-4.1-nano	0.700	0.660	1.000	1.000	0.340
Partial observability	gpt-5.5	0.740	0.653	1.000	1.000	0.347

Table 4: Cross-model held-out listener audit. GPT-5.5 rows reuse the existing speaker candidates and selected messages; the oracle is 1.000 because population-play already selects an all-listener-successful candidate in every scene.

Finding 4: coordinate fallbacks are useful, but selector quality matters without them. The strongest population-play result uses candidate sets that include an explicit row/column fallback. Table 5 asks whether the result is merely a coordinate fallback effect. We remove candidate messages that explicitly mention row or column numbers and recompute selection without new API calls. This ablation is deliberately stricter and exposes a limitation of the current local population selector.

Robust non-coordinate candidates remain available: the no-coordinate oracle reaches 0.793, and consensus+info reaches 0.760. However, naive population-play falls to 0.420 because the local training listeners often assign equal or low scores to more informative non-coordinate descriptions, causing the length penalty to favor short ambiguous messages. Thus the headline population-play result should be read as robust fallback selection when explicit listener-invariant coordinates are available, while the ablation shows that cross-play is also useful for diagnosing selector weaknesses. The same cache-only filter on the mixed pilot gives a no-coordinate oracle of 0.960 and consensus+info success of 0.920, with gains over mirror concentrated in relational-reference and perspective-shift scenes.

Method	Cross-play	Same-play	Gap	Tokens
Direct first	0.553	-	-	11.7
Best-of-K shortest	0.367	-	-	4.6
Mirror self-play	0.580	1.000	0.420	8.3
Population-play	0.420	0.520	0.100	5.0
Consensus+info	0.760	-	-	12.5
Informativeness prior	0.760	-	-	12.5
Oracle candidate	0.793	-	-	9.4

Table 5: No-exact-coordinate ablation on the alternate-model audit. We remove candidate messages that explicitly mention row or column numbers, then recompute selection while reusing cached held-out listener evaluations.

Selection behavior explains the ablation. A candidate-role audit shows that, in the full perspective-stress and partial-observability runs, population-play selects the coordinate-fallback slot in 1.000 of scenes; in the alternate-model perspective audit, this is 50 coordinate selections versus 17 for mirror self-play. Once exact coordinates are removed, naive population-play selects a spatially informative message in only 4 of 50 perspective-stress scenes and drops to 0.420 cross-play success. Consensus+info falls back to the informativeness prior in all 50 stress scenes, selecting spatially informative messages throughout and

reaching 0.760. In the mixed no-coordinate pilot, consensus+info uses local consensus in 32 scenes and the informativeness fallback in 18 scenes.

Setting	Gap	Same-play minus cross-play
Mixed 50, mirror	0.093	████
Mixed 50, population	0.000	
Perspective stress, mirror	0.187	████████
Perspective stress, population	0.000	

Figure 1: Cross-play gap for mirror and population selection. The gap is same-play success minus held-out cross-play success.

Failure decomposition: generation versus selection. The oracle result is important. In the perspective-stress replication, the oracle candidate succeeds on every scene, so the API speaker usually generated a robust message. The failure is selection: mirror self-play often selects a shorter message that is only partner-robust, whereas population-play selects the explicit fallback or a less frame-sensitive description.

A cache-only selection-regret audit makes this decomposition explicit. Regret is oracle candidate success minus method success, averaged by scene. In the full-candidate API runs, population-play has zero regret in the mixed, perspective-stress, alternate-model, and partial-observability settings, while mirror self-play leaves regrets of 0.093, 0.187, 0.287, and 0.333 respectively. In the no-coordinate ablations, consensus+info reduces regret to 0.040 in the mixed pilot, 0.033 in the perspective-stress audit, and 0.013 in partial observability, showing that most remaining failures are selector failures rather than missing candidates.

An even stricter candidate-pool audit asks whether any candidate succeeds with every held-out listener in a scene. Every full-candidate run has at least one such robust candidate in all 50 scenes, and population-play selects a robust candidate in 1.000 of scenes. Mirror robust-selection rates are 0.860, 0.720, 0.560, and 0.500. After exact coordinates are removed, robust non-coordinate candidates remain available in 0.920, 0.660, and 1.000 of scenes for the mixed, alternate-model perspective, and partial-observability runs, and consensus+info matches the per-scene oracle in 0.940, 0.920, and 0.980 of scenes.

Ruling out simple alternative explanations. A uniform-random candidate baseline rules out the explanation that any candidate would do. Random expected success is 0.865, 0.715, 0.657, and 0.753 in the four full-candidate runs, so population-play improves over random by 0.135, 0.285, 0.343, and 0.247. In the no-coordinate runs, consensus+info improves over random by 0.100, 0.218, and 0.316.

A message-length audit rules out a pure verbosity explanation. Population-play does select longer messages than mirror self-play by 1.26, 2.82, 2.82, and 3.16 whitespace tokens on average in the four full-candidate runs, but direct messages are longer than mirror in the perspective and partial-observability runs while remaining less successful. Population-play messages also remain near the deterministic template scale: 6.82, 10.22, 10.22, and 9.66 tokens on average, compared with a 10-token template baseline.

A prefix-K candidate-budget audit shows why the four-candidate protocol is useful. In the alternate-model perspective audit, robust-scene coverage rises from 0.380 at $K = 1$ to 0.660 at $K = 3$ and 1.000 at $K = 4$; in partial observability, it rises from 0.620 at $K = 1$ to 1.000 at $K = 3$. Candidate diversity matters, but only if the selector can identify the robust candidate.

A held-out listener-disagreement audit shows that mirror self-play produces less stable messages. Mirror self-play yields split held-out outcomes in 0.080, 0.180, 0.360, and 0.500 of scenes across the full-candidate runs, while population-play has 0.000 split outcomes in all four. In no-coordinate partial observability, consensus+info reduces split outcomes from mirror’s 0.540 to 0.020. A listener-confidence audit further shows that self-reported

uncertainty is not enough: in the alternate-model perspective runs, ambiguity flags are 0.000 while mirror self-play still produces high-confidence failures (confidence ≥ 0.8) in 0.147 of full-candidate listener rows and 0.280 of no-coordinate listener rows.

API-listener leave-one-out mechanism check. To check that the result is not only an artifact of deterministic local selectors, we also run a cache-only leave-one-listener-out analysis over the API all-candidate evaluations. For each scene, one API listener prompt is held out; API-mirror selection uses one different API listener prompt, while API-population selection uses the other two. Population selection improves over API-mirror selection in all four cached runs. The gains are small in the mixed and main perspective runs, but larger for the alternate-model perspective audit (0.940 vs. 0.847) and partial observability (0.940 vs. 0.767). Because this analysis reuses the cached API listener pool for post-hoc splits, we treat it as a mechanism check rather than as the main held-out evaluation.

Partial-observability support check. As an additional support check, we run a 50-scene partial-observability stress setting in which the speaker sees a private landmark hidden from the listener. The prompt instructs the speaker to use only listener-visible information. In this setting, mirror self-play obtains 0.667 cross-play success despite 1.000 same-play success, while population-play and the oracle both reach 1.000. After removing exact row/column candidates, robust non-coordinate messages remain available: the no-coordinate oracle is 1.000 and consensus+info reaches 0.987. The candidate audit finds zero explicit private-landmark references, so the failures reflect under-informative messages that fit both the target and distractor rather than object-ID leakage or hidden-landmark leakage. Rubric coding supports this interpretation: all 50 full-run mirror failures and all 59 no-coordinate mirror failures are underspecified-distractor choices.

7 Qualitative Analysis

The quantitative pattern is easy to see in individual scenes. In scene ps_000005, the target is a small green sphere at row 4, column 1, and there is another small green sphere at row 1, column 4. Mirror self-play selects “It’s a small green sphere.” Its local partner resolves this correctly, but two of the three alternate-model held-out listeners choose the distractor. Population-play selects “The target is the small green sphere at row 4, column 1,” which all held-out listeners map to the target. The mirror message is not syntactically wrong; it is pragmatically under-informative for a listener who sees two compatible objects.

In scene ps_000011, mirror self-play selects “Find the large yellow cube left of the purple cylinder.” Under perspective shift, this relation is frame-sensitive: one alternate-model held-out listener chooses the same-shape, same-color distractor. Population-play selects “Go to row 4, column 1: the large yellow cube,” which all held-out listeners map to the target. Here the failure is not merely brevity; the problem is that the selected relation depends on an orientation convention that is not shared by all listeners.

These examples suggest two ways in which same-partner success can mislead. First, the training partner may resolve an ambiguous attribute description in the target’s favor because of a deterministic tie-breaker. Second, the training partner may share or assume the speaker’s spatial frame. Both cases produce perfect same-play success without robust communicative success. Population-play helps when at least one local listener penalizes the ambiguity or when the candidate pool contains a listener-invariant fallback.

The rubric-coded mirror failures used in the analysis show a concentrated failure mode. Across 152 listener-level failures, 147 are underspecified distractor cases and 5 are perspective-frame errors; no coded failures are wrong-attribute, private-landmark, or listener-misparsed cases. A generated qualitative appendix adds two partial-observability examples, including a no-coordinate consensus+info repair that raises held-out success from 0.333 to 1.000 in the selected scene.

8 Discussion

The experiments support a modest but important conclusion: communicative success is not a sufficient evaluation signal unless the partner distribution is specified. In PRAG-CrossPlay, mirror self-play is not failing because it cannot find any working message. It is failing because a message that works with one partner can be under-specified for another. Population-play improves robustness by making the selector satisfy multiple listener assumptions, but the no-coordinate ablation shows that population diversity must be paired with a selector that values discriminative information.

This has implications for interactive language-agent research. Many agent pipelines include a step that resembles our selection loop: generate alternatives, simulate a user or partner, and keep the response that appears to succeed. If evaluation uses the same partner type, the resulting score may reward private conventions, hidden tie-breakers, or brittle spatial assumptions. Cross-play offers a low-cost diagnostic: hold out partner prompts, model families, or humans, and report the same-play/cross-play gap. A large gap does not necessarily mean the agent is useless; it identifies where the communication protocol has become partner-specific.

The no-coordinate ablation is also instructive. Exact coordinates are useful in our text-rendered grid because they are listener-invariant. In more natural embodied settings, such fallbacks may be unavailable, unnatural, or costly. The relevant lesson is not that agents should always use coordinates, but that robust communication often requires grounding descriptions in information that the listener can reliably observe and interpret. When such descriptions are unavailable or not selected, cross-play exposes the weakness.

9 Limitations

The current experiments are intentionally small and diagnostic. The selector partners are local deterministic listeners rather than trained LLM agents. This is useful for isolating partner overfitting, but it means the result should be framed as a cross-play diagnostic rather than as evidence about large-scale agent training. The held-out listeners are API language models, not humans. We audit three held-out model families including gpt-5.5, but they are still language-model listeners and prompt variants rather than independent human partners. The strongest population-play result relies on exact coordinate fallbacks; without those fallbacks, the current local listener population is not sufficient, although consensus+info recovers much of the no-coordinate oracle. The grid scenes are text-rendered, so the benchmark tests situated reference rather than embodied perception. Finally, the strongest evidence is concentrated in two diagnostic families, perspective shift and partial observability; broader claims require larger balanced runs across all scenario types and, ideally, human listener validation.

10 Conclusion

Same-partner communicative success is not sufficient evidence of robust pragmatic communication. In PRAG-CrossPlay, mirror self-play can achieve perfect same-play success while failing under held-out listeners, especially when messages are perspective-sensitive or under-specified. Population-play mitigates this failure when listener-invariant fallbacks are available, but the no-coordinate ablation shows that cross-play is also useful for diagnosing and repairing weaknesses in the selector itself. These results support cross-play as a simple, objective evaluation axis for language agents that learn, filter, or select messages through situated interaction.

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